



Responsible AI in Social Media Recommendation Systems

Filter Bubbles · Bias · Explainability & Privacy in Algorithmic Feeds

Dataset: Twitter/Reddit Engagement Dataset (Kaggle) | **Tools:** Python · Scikit-learn · SHAP · Fairlearn · PySyft

What is a Social Media Recommender?

Social media platforms use AI-driven recommendation engines to decide what content appears in each user's feed, trending section, or For You page. These systems optimize for engagement metrics — likes, shares, watch time — using collaborative filtering, graph neural networks, and transformer-based ranking models.

The ethical problem: Maximizing engagement $\neq$ maximizing well-being. These systems can radicalize, polarize, and discriminate at societal scale.

3.8B+

Daily active social
media users

Global reach

70%

Facebook feed
controlled by
algorithms

Content distribution

40%

Twitter users led to
extreme content

Engagement bias

Relevance to Responsible AI: Algorithmic feeds shape public opinion, mental health, and democratic discourse — demanding fairness, transparency, and accountability.



How Social Media Recommendation Systems Work

01

Collaborative Filtering

User-item interaction matrices

02

Content-Based Filtering

NLP + CV feature matching

03

Graph Neural Networks

Social graph relationships

04

Two-Tower Models

TikTok/YouTube large-scale retrieval

05

Reinforcement Learning

Long-term engagement sequences

Real-World Ethical Cases

Facebook (2021)

Algorithm amplified divisive political content (WSJ Files)

TikTok FYP (2020)

Racial & political bias in content suppression (lawsuit)

YouTube (2019)

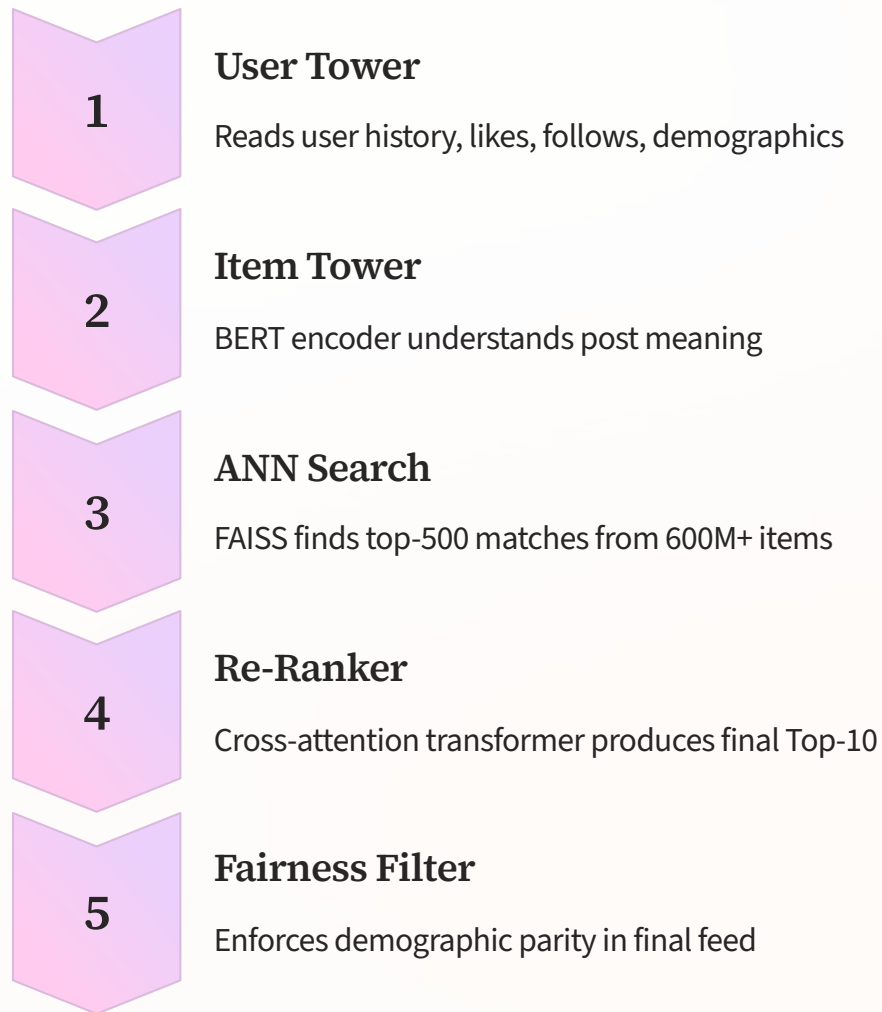
Rabbit-hole algorithm drove radicalization (Ribeiro et al.)

Twitter/X (2021)

Right-leaning content amplified by algorithm (audit)

Two-Tower Transformer Architecture

The Two-Tower Transformer is the dominant model architecture used by TikTok, YouTube, Twitter, and LinkedIn for large-scale content ranking. It gets its name from having two separate neural networks (towers) that work in parallel — one for the user, one for the content.



Why Two-Tower? A naive approach would compare every user against every post — but YouTube has 800 million videos and 50 million daily active users. That is 40 quadrillion comparisons per day. The Two-Tower model solves this by pre-computing item vectors offline and only computing the user vector in real-time, finding matches in milliseconds instead of days.

Ethical Risks in Recommendation Systems



Bias & Discrimination

- Demographic bias: race/gender underrepresented in trends
- Political bias: amplification of ideologically extreme content
- Popularity bias: viral posts crowd out quality niche content



Privacy Violations

- Micro-profiling from scroll patterns, dwell time, mouse hover
- Cross-platform data linkage re-identifies anonymous users
- Shadow profiles built without user knowledge or consent



Filter Bubbles & Echo Chambers

- Engagement-maximizing feeds reinforce existing beliefs
- Reduces exposure to diverse viewpoints and perspectives
- Documented radicalization pathways (YouTube, Facebook)



Lack of Transparency

- Users cannot understand why content is shown or hidden
- Opaque content moderation inconsistently applied
- GDPR Article 22: right to explanation not fulfilled



Responsible AI Techniques



Fairness Metrics

- Demographic Parity across user demographic groups
- Equalized Opportunity (EO) for content exposure rates
- GINI Coefficient for recommendation diversity
- Counterfactual Fairness via do-calculus interventions



Explainability Tools

- SHAP values: per-user feature attribution for feed decisions
- Attention weight visualization on Transformer ranker
- Counterfactual explanations: 'Why not show X instead?'
- Global surrogate models (LIME) for policy-level audits



Bias Mitigation

- Adversarial debiasing: penalize demographic-predictive features
- Post-hoc re-ranking: enforce group representation in Top-K
- Calibrated diversity regularization during training
- Data augmentation for underrepresented content groups



Privacy Techniques

- Differential Privacy (epsilon-DP) on gradient updates
- Federated Learning: train on-device, share only gradients
- K-anonymity grouping for user interest profiles
- Secure Aggregation: cryptographic gradient aggregation

Dataset & Analysis Plan

Twitter Sentiment & Engagement Dataset (Kaggle)

~1.6M labeled tweets

Positive, negative, neutral sentiment labels

User engagement data

User IDs, likes, retweets, replies, follower count

Content metadata

Hashtags, topics, timestamps, text, media type, URLs

Supplementary data

Reddit r/news dataset (Pushshift API)

5-Step Analysis Plan

- Step 1 — EDA:** Explore engagement patterns; identify demographic disparities
- Step 2 — Baseline RS:** Build CF + content-based; benchmark NDCG, CTR, Recall@K
- Step 3 — Bias Audit:** Measure Demographic Parity & EO gaps
- Step 4 — Fairness Fix:** Re-ranking + adversarial debiasing; measure delta accuracy vs delta fairness
- Step 5 — XAI + DP:** SHAP explanations + epsilon-DP experiments; privacy trade-off curve

Trade-offs: Accuracy vs Responsibility

Implementing responsible AI techniques involves balancing competing objectives. The key insight is that **accuracy and responsibility are not mutually exclusive** — **they are complementary goals** that can be optimized together.

Accuracy vs Fairness

Baseline: 88% NDCG@10

With Responsible AI: +DP: 82% | +DP+EO:
75%

Best Setting: $\lambda=0.3$ Pareto-optimal

Fairness re-ranking reduces NDCG ~15%.
Deep models gain ~15% CTR but need more data.

Accuracy vs Privacy

Baseline: $\epsilon=0.1$: 48% | $\epsilon=1$: 65%

With Responsible AI: FedLearn: 82%

Best Setting: $\epsilon=1-5$ FedLearn preferred

DP at $\epsilon=0.1$ degrades Recall ~20%. Federated Learning preferred.

Accuracy vs Explainability

Baseline: SVD: 65% | LightFM: 78%

With Responsible AI: 2-Tower: 88%

Best Setting: SHAP post-hoc negligible latency

Deep models gain ~15% CTR but need more data. SHAP adds minimal overhead.

Core Insight: Accuracy and Responsibility Are Complementary

Social media recommendation systems are among the most ethically consequential AI deployments in existence. By rigorously applying fairness metrics, SHAP-based explainability, and differential privacy — and grounding each technique in peer-reviewed IEEE and Springer research — this project demonstrates that **accuracy and responsibility are not mutually exclusive: they are complementary goals.**

Dataset

Twitter Sentiment (Kaggle)

Tools

Python · Fairlearn · SHAP · PySyft

Research

IEEE Access · IEEE TKDE · IEEE TOE · IEEE TIFS ·
Springer AI&Ethics · UMUI

Thank You

Questions & Discussion